ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

M. Tech. / B. Tech. M. Tech. (CSE)

Semester: II/VIII

SUBJECT CODE: MCS213

NETWORK SECURITY AND CRYPTOGRAPHY

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is the difference between Cryptography and steganography?

(ii) Distinguish between Active and Passive attacks.

(iii) Define block cipher.

(iv) Define hash algorithm.

(v) Define birthday attack.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define Authentication.

(ii) What is the Symmetric Key Cryptography?

(iii) What do you mean by e-mail security?

(iv) What is IP security?

(v) What do you mean by nonce?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Compare and contrast substitution cipher with transposition cipher in the terms of Classical encryption techniques.

OR

(b) Differentiate between hash Codes and Message authentication codes (MAC).

(ii)(a) What are the examples of Firewalls? Explain them.

OR

(b) What are the requirements for the use of public key Certificates schemes?

Q.4.(i)(a) Write the objectives of HMAC. Describe the HMAC algorithm.

OR

(b) Determine the GCD of (24561,17892) using Euclid algorithm.

(ii)(a) Find the secret key shared between user A and B using Diffie Hellman Algorithm for the following.

$q = 353$, $\alpha = 3$, $XA = 45$, $XB = 50$

OR

(b) Explain Secure Electronic Transaction (SET) protocol with its component.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Explain the concept of Digital Signature algorithm with key generation and verification in detail.

OR

(b) Explain briefly about the architecture and certification mechanism in Kerberos.

Q.6.(a) Draw the block diagram of DES function and explain the strength of DES.

OR

(b) Describe the role of Encapsulating Security Payloads (ESP) in terms of IP Security.